DEBRIEF · SIMULATED CASE

Class IV haemorrhage, PEA and resuscitation

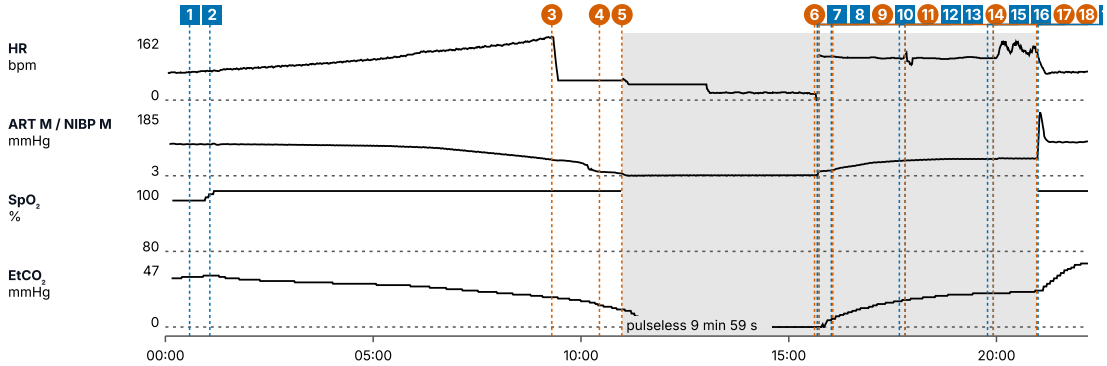
M 35 y 75 kg · 22:12 simulated (5 min 10 s on the clock) · 10/24/2026, 1:14:50 AM · Run by: Live demo, 28 October

Key times

Start → first team action	Start high-quality CPR without delay (debrief: within 30 s o…	Give volume and epinephrine; recognise the return of…	Pulseless in all
35 s	**4 min 42 s**	**9 min 59 s**	**9 min 59 s**
00:00 → 00:35 (Bleeding, oxygen by face mask (FiO_2 1.0))	10:59 → 15:41 · asked ≤ 30 s	10:59 → 20:58 · asked ≤ 6 min 00 s	first lost 10:59

The course

■ team action · ● patient event · □ scenario state · numbers match Key moments



Objectives

What the record shows: it reads the run, it does not grade the person

1. – Not assessable Recognise PEA from hypovolaemia
   This objective has no rule: the instructor judges it.
2. **✕ Not observed** Start high-quality CPR without delay (debrief: within 30 s of the pulse loss — draft, owner to confirm) 4 min 42 s of 30 s
   CPR 4 min 42 s after the pulse loss (10:59 → 15:41; within 30 s asked).
3. **✕ Not observed** Give volume and epinephrine; recognise the return of circulation on the arterial line and capnogram (debrief: the pulse back within 6 min of its loss — draft, owner to confirm) 9 min 59 s of 6 min 00 s
   The return of circulation 9 min 59 s after the pulse loss (10:59 → 20:58; within 6 min 00 s asked).

Key moments and what the monitor showed

values at that second

n	Time	Who	What happened	HR	MAP	SpO_2	$EtCO_2$
1	00:35	Team	Bleeding, oxygen by face mask (FiO_2 1.0)	71	93	96	37
2	01:04	Team	Bleeding, intubated and ventilated (12 × 500, PEEP 5, FiO_2 1.0)	74	93	98	38
3	09:17	Patient	Pre-arrest: junctional rhythm 50/min	160	50	99	22
	Alarms since the last row: $EtCO_2$ low, SpO_2: low perfusion, ART S low, ART M low, HR high						
4	10:26	Patient	PEA: junctional rhythm, no pulse	50	15	99	16
	Alarms since the last row: ART D low, ART: no pulsatile pressure, ART S low, ART D low						
5	10:59	Patient	Pulse lost (the model)	50	9	–	13
	Alarms since the last row: ART: no pulsatile pressure, SpO_2: low perfusion						
6	15:37	Patient	Asystole	16	5	–	0
	Alarms since the last row: HR low, SpO_2: no pulse detected, Apnoea (no CO_2 breaths)						
7	15:41	Team	CPR: compressions 110/min, bag via the tube 10/min, FiO_2 1.0	0	4	–	0
	Alarms since the last row: Asystole						
8	15:41	Team	CPR: 2 L warmed crystalloid and 2 units of red cells running	0	4	–	0
9	15:43	Patient	CPR continuing	114	14	–	0
10	16:01	Team	CPR: epinephrine 1 mg given	111	19	–	5
	Alarms since the last row: ART S low, ART D low, awRR low, $EtCO_2$ low, SpO_2: low perfusion						
11	16:03	Patient	CPR continuing	111	19	–	6
12	17:39	Team	Pulse check: VF	107	46	–	19

7 more in the timeline

SpO_2 –: no reliable saturation reading without a pulse; a pleth signal during CPR reflects compression flow, not oxygenation.

Measured facts

Lowest SpO_2	96 % at 00:10
Lowest ART M / NIBP M	3 mmHg at 11:11
Highest $EtCO_2$	47 mmHg at 22:02
Highest T1 / T2	36.8 °C at 00:01
Apnoea alarm to ventilated breaths	none after the alarm at 11:51
Pulseless	9 min 59 s in all; first lost at 10:59
Return of circulation	9 min 59 s after the pulse loss (20:58)
Epinephrine, boluses in all	1 mg (1 dose)
Labs (the model): last (lowest–highest)	K^+ 4.4 mmol/L (4.2–4.5) · Lactate 4.7 mmol/L (1.0–4.7) · pH 7.18 (7.13–7.41) · Glucose 7.5 mmol/L (5.6–7.5) · Tcore 36.2 °C (36.1–36.8)

Talking points

Reactions	Description	Analysis	Summary
How did that feel? What stands out for you?	In a sentence or two: what was the problem, and when did you know?	Take one key moment: what did you see, what did you think, what did you do?	One thing to keep doing, one thing to do differently next time.

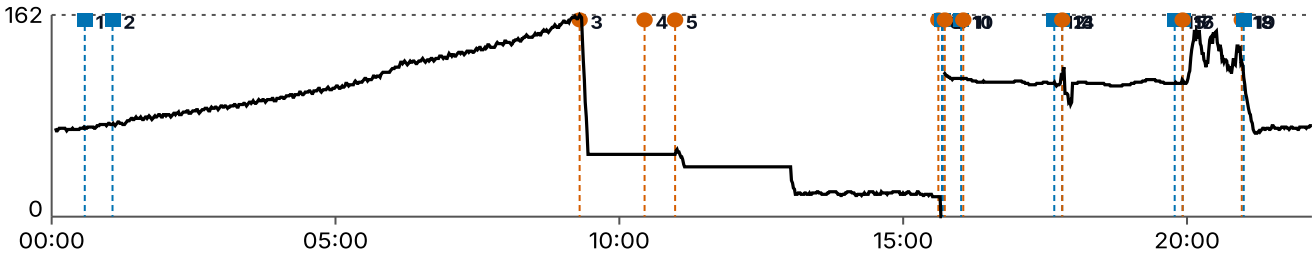
Notes

Kit round A4 rehearsal of the live-demo path.

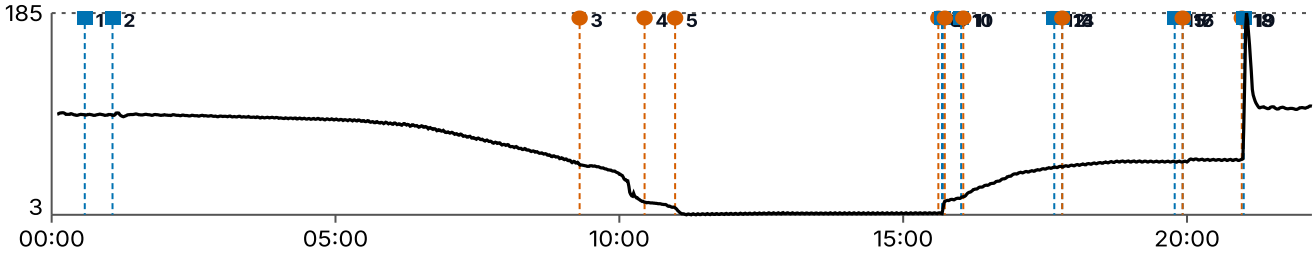
Trends

■ team action · ● patient event · | scenario state

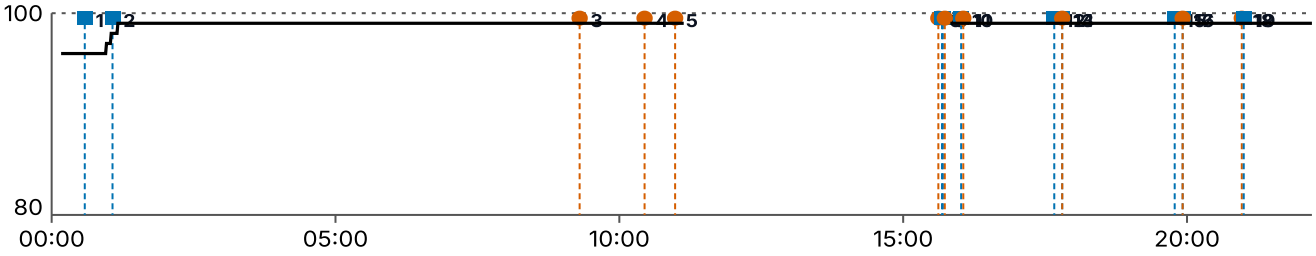
HR bpm



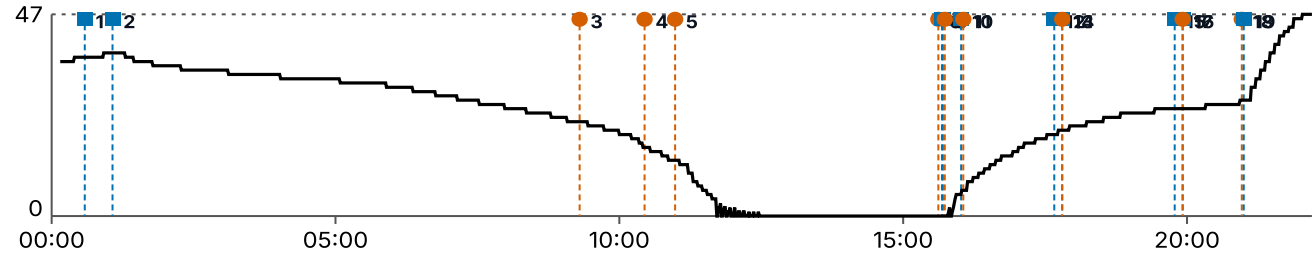
ART M / NIBP M mmHg



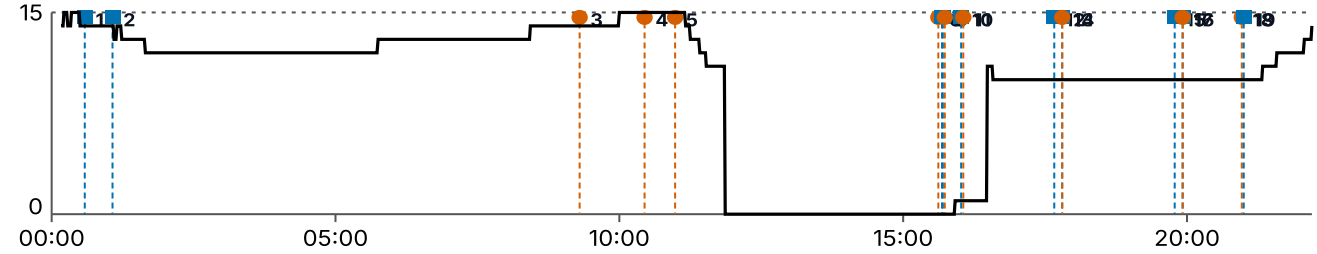
SpO_2 %



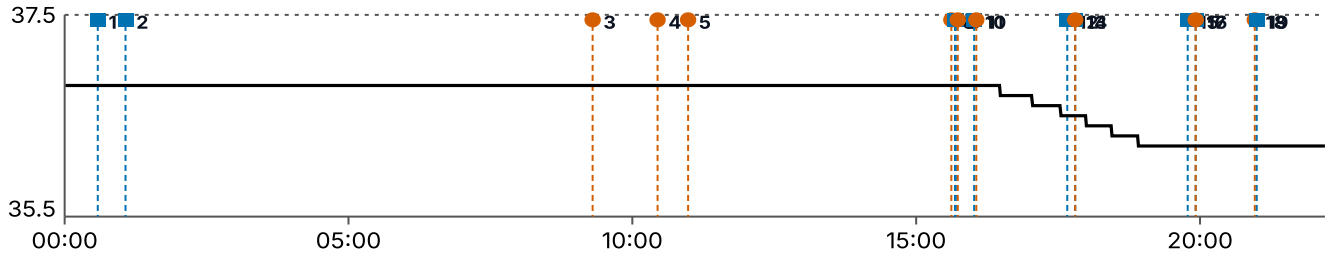
$EtCO_2$ mmHg



awRR / RR /min



T1 / T2 °C



Drugs and fluids

Time	Given
15:41	Crystalloid 2000 mL over 3 min
15:41	Blood 600 mL over 5 min
16:01	Epinephrine 1 mg IV

- Epinephrine: 1 mg in 1 bolus

Timeline

Time	Kind	Event
00:00	Scenario	Scenario loaded: Class IV haemorrhage, PEA and resuscitation
00:00	Scenario	Scenario state: Bleeding 2.5 L over 10 min, breathing room air
00:00	Scenario	Blood loss 2500 mL over 10 min
00:01	Action	Speed ×8
00:04	Bookmark	Bookmark: Bookmark 1 at 00:04
00:35	Scenario	Scenario state: Bleeding, oxygen by face mask (FiO_2 1.0)
00:35	Scenario	Preoxygenation, FiO_2 100 %
01:04	Scenario	Scenario state: Bleeding, intubated and ventilated (12 × 500, PEEP 5, FiO_2 1.0)
01:04	Scenario	Tracheal tube in place
01:04	Scenario	Preoxygenation, FiO_2 21 %
01:04	Scenario	Ventilator: RR 12 /min, VT 500 mL, PEEP 5 cmH_2O, FiO_2 100 %
01:04	Patient	Ventilated breaths began
06:23	Alarm	$EtCO_2$ low
08:21	Alarm	SpO_2: low perfusion
08:27	Alarm	ART S low
08:36	Alarm	ART M low
08:49	Alarm	HR high
09:17	Scenario	Scenario state: Pre-arrest: junctional rhythm 50/min
09:18	Scenario	Rhythm: Junctional escape
09:20	Action	Speed ×4
10:05	Alarm	ART D low
10:11	Alarm	ART: no pulsatile pressure

Time	Kind	Event
10:16	Alarm	ART S low
10:16	Alarm	ART D low
10:26	Scenario	Scenario state: PEA: junctional rhythm, no pulse
10:50	Alarm	ART: no pulsatile pressure
10:57	Alarm	SpO_2: low perfusion
10:59	Patient	Pulse lost (the model)
10:59	Action	Speed ×8
11:06	Alarm	HR low
11:09	Alarm	SpO_2: no pulse detected
11:51	Alarm	Apnoea (no CO_2 breaths)
15:37	Scenario	Scenario state: Asystole
15:38	Action	Speed ×1
15:39	Alarm	Asystole
15:41	Scenario	Scenario state: CPR: compressions 110/min, bag via the tube 10/min, FiO_2 1.0
15:41	Scenario	CPR started, 110 /min
15:41	Scenario	Bag-mask: RR 10 /min, VT 500 mL, PEEP 5 cmH_2O, FiO_2 100 %
15:41	Scenario	Scenario state: CPR: 2 L warmed crystalloid and 2 units of red cells running
15:41	Scenario	Crystalloid 2000 mL over 3 min
15:41	Scenario	Blood 600 mL over 5 min
15:43	Scenario	Scenario state: CPR continuing
15:47	Alarm	ART S low
15:47	Alarm	ART D low
15:55	Alarm	awRR low
15:55	Alarm	$EtCO_2$ low
15:57	Alarm	SpO_2: low perfusion
16:01	Scenario	Scenario state: CPR: epinephrine 1 mg given
16:01	Scenario	Epinephrine 1 mg IV
16:02	Action	Speed ×4
16:03	Scenario	Scenario state: CPR continuing
17:39	Action	Speed ×1
17:39	Scenario	Scenario state: Pulse check: VF
17:39	Scenario	Rhythm: Coarse VF
17:40	Action	Defibrillator charging to 200 J
17:42	Alarm	Ventricular fibrillation or tachycardia
17:48	Action	Shock delivered
17:48	Scenario	Scenario state: Shocked: CPR on for 2 min
17:48	Action	Speed ×4
19:46	Action	Speed ×1
19:46	Scenario	Scenario state: Pulse check: VF again
19:46	Scenario	Rhythm: Coarse VF

Time	Kind	Event
19:48	Action	Defibrillator charging to 200 J
19:49	Alarm	Ventricular fibrillation or tachycardia
19:55	Action	Shock delivered
19:55	Scenario	Scenario state: Shocked again: CPR on
20:58	Patient	Circulation returned (the model)
21:00	Scenario	Scenario state: Return of circulation: CPR stopped, ventilator 12 × 500, PEEP 5, FiO_2 1.0
21:00	Scenario	CPR stopped
21:00	Scenario	Ventilator: RR 12 /min, VT 500 mL, PEEP 5 cmH_2O, FiO_2 100 %
21:00	Action	Speed ×4
21:02	Alarm	ART S high
21:02	Alarm	ART M high
21:03	Alarm	ART D high

Bookmarks

- 00:04 Bookmark: Bookmark 1 at 00:04

Appendix: one row per minute

Time	HR bpm	ART M / NIBP M mmHg	SpO_2 %	$EtCO_2$ mmHg	awRR / RR /min	T1 / T2 °C	K^+ mmol/L	Lactate mmol/L	pH	Glucose mmol/L	Tcore °C
00:00	71	94	96	36	14	36.8	4.2	1.0	7.41	5.6	36.8
01:00	75	94	97	38	14	36.8	4.2	1.0	7.39	5.6	36.8
02:00	82	93	99	35	12	36.8	4.2	1.0	7.40	5.6	36.8
03:00	89	92	99	34	12	36.8	4.2	1.1	7.40	5.6	36.8
04:00	95	90	99	33	12	36.8	4.2	1.1	7.40	5.6	36.8
05:00	104	89	99	32	12	36.8	4.2	1.2	7.39	5.6	36.8
06:00	119	85	99	30	13	36.8	4.2	1.3	7.39	5.6	36.8
07:00	131	77	99	28	13	36.8	4.2	1.5	7.39	5.6	36.8
08:00	141	66	99	25	13	36.8	4.2	1.7	7.38	5.6	36.8
09:00	156	54	99	23	14	36.8	4.2	1.9	7.37	5.6	36.8
10:00	50	40	99	19	15	36.8	4.2	2.1	7.36	5.6	36.8
11:00	53	9	99	13	15	36.8	4.2	2.3	7.35	5.6	36.8
12:00	40	4	–	0	0	36.8	4.2	2.6	7.31	5.7	36.8
13:00	40	4	–	0	0	36.8	4.2	2.9	7.28	5.8	36.8
14:00	19	4	–	0	0	36.8	4.2	3.2	7.26	5.9	36.8
15:00	19	4	99	0	0	36.8	4.2	3.6	7.23	6.1	36.8
16:00	111	18	99	5	1	36.8	4.2	3.8	7.21	6.2	36.7
17:00	109	40	99	15	10	36.7	4.3	4.0	7.20	6.4	36.5
18:00	107	47	99	21	10	36.4	4.4	4.1	7.17	6.6	36.3
19:00	106	51	99	24	10	36.2	4.4	4.3	7.15	6.9	36.1
20:00	108	51	99	25	10	36.2	4.4	4.5	7.14	7.1	36.1
21:00	112	81	99	27	10	36.2	4.4	4.7	7.13	7.3	36.2
22:00	72	100	99	46	12	36.2	4.4	4.7	7.16	7.5	36.2